Local adaptation of the rate of acclimation in the freshwater copepod *Skistodiaptomus pallidus*

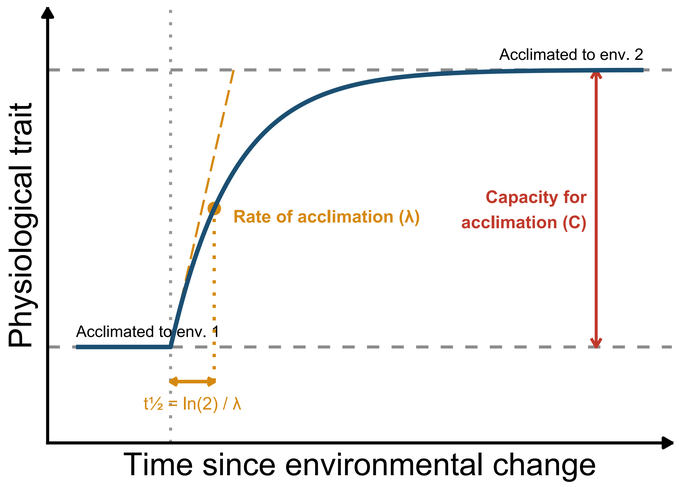
The Zooplankton Ecology & Adaptation Lab At UMass Lowell

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Introduction

- Heatwaves are projected to increase in duration and intensity by the end of the century
- Acclimation: Short-term and often reversible change in physiology
  - No genetic change required
- Temporal dynamics of acclimation defined by **capacity** and **rate of acclimation**



- Minimal information on how these parameters vary across populations

Central Question

Do acclimation parameters vary across populations?

Hypotheses

1. The acclimation capacity will be higher in the population with a higher average surface temperature.
2. The acclimation capacity and rate of acclimation will have a negative relationship.

Results

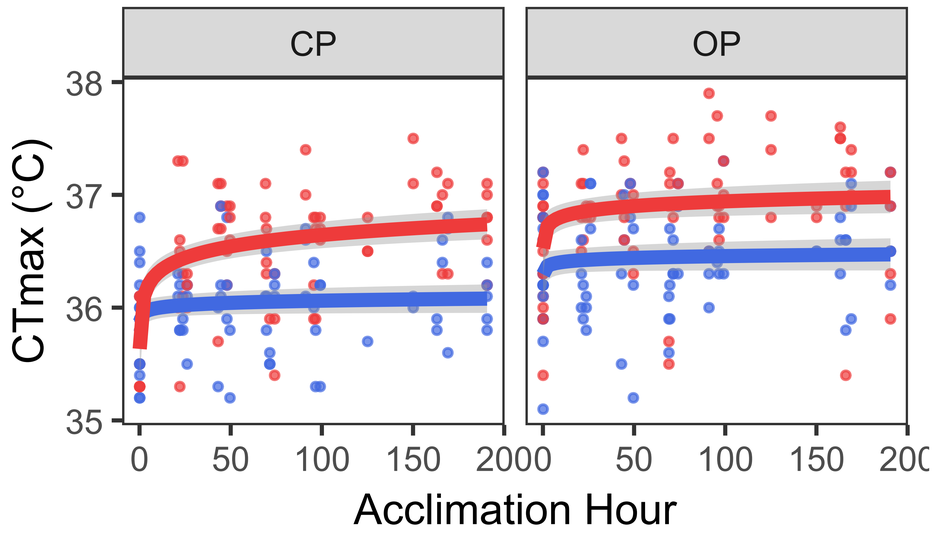


Figure 1: CTmax (°C) over time (Acclimation hours) for the control (blue) and warming (red) treatments. The two populations are shown separately. Each point represents an individual copepods upper thermal limit. Logistic regressions are fit to each treatment group.

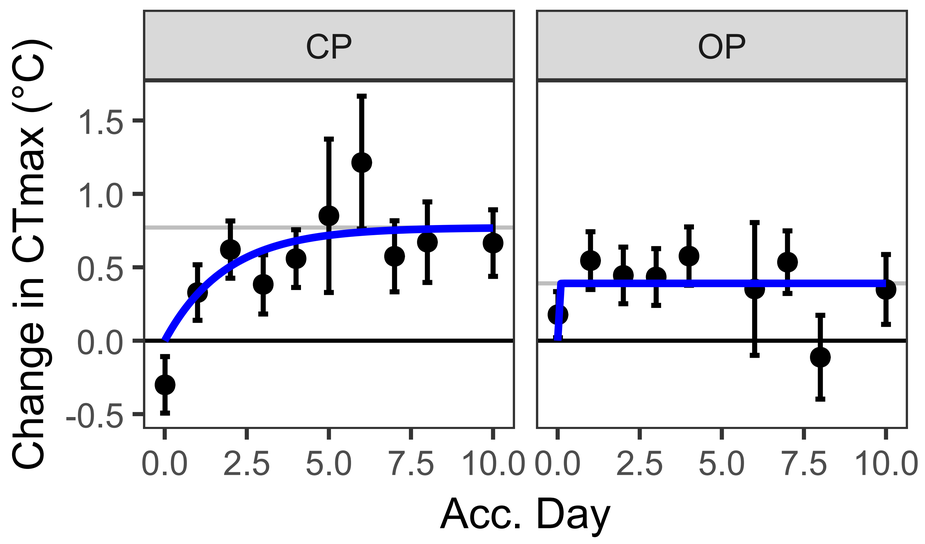
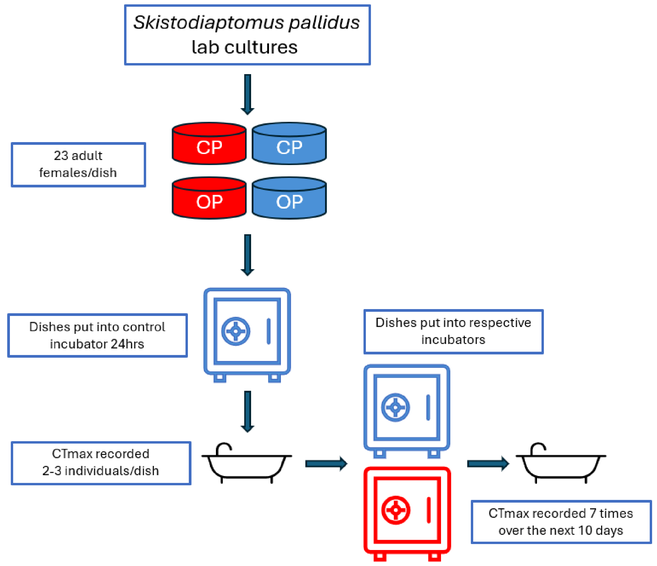


Figure 2: Plot of the model contrasts (Warming - Control; °C) over time (Acclimation Day) for the two populations. The blue lines indicate the fitted acclimation equation based on the rate and capacity parameters.

Methods



Conclusions

- *S. pallidus* is able to acclimate rapidly to warming.
- Populations differ in both capacity and acclimation, with a potential negative correlation between capacity and rate of acclimation.

Next Steps

- Replicate with additional populations of *S. pallidus*
- Characterizing lake temperature dynamics to examine environmental drivers of the evolution of acclimation parameters.

Acknowledgements

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